Nike Retail Sales Performance Analysis Using SQL

Prepared by

Akash

Tools Used: PostgreSQL, Microsoft Excel

Project Overview

Retail businesses generate large volumes of sales transactions across multiple locations, products, and retailers. Analysing this data manually is inefficient and often leads to poor business decisions.

This project analyses Nike retail sales data using SQL to identify patterns in product performance, retailer contribution, regional demand, and profitability. The raw dataset is transformed into a normalized relational database structure to support efficient analysis and business intelligence.

Using SQL techniques such as joins, aggregations, common table expressions, and window functions, the project extracts meaningful insights that help understand sales trends and operational performance.

The goal is to convert raw retail transaction data into actionable insights that support data driven decision making.

Problem Statement

Retail sales data often exists in a raw and unstructured format. This creates several challenges.

Data redundancy due to repeated retailer, product, and location information.
Difficulty identifying high performing products and retailers.
Lack of visibility into regional sales patterns.
Limited ability to evaluate profitability and operational efficiency.

Without proper data modelling and analysis, businesses struggle to optimize pricing, inventory management, and sales strategy.

This project addresses these challenges by cleaning the dataset, normalizing the database structure, and performing SQL based business analysis.

SALES PERFORMANCE ANALYSIS

I. Current Situation

• The business currently stores Nike retail sales data in a single raw dataset that contains retailer information, product details, location data, and transaction records in one table. This structure makes detailed analysis difficult.

• Retailer names and IDs are repeatedly stored across thousands of transactions, creating unnecessary data redundancy.

• Product names, state, and city information appear multiple times across records, increasing the risk of inconsistent or incorrect values.

• Important financial fields such as operating margin are stored as percentage text rather than numeric values, which makes calculations and analysis more difficult.

• Because all data exists in one table, identifying top performing retailers, profitable products, and high demand regions requires complex manual querying.

• Sales trends across quarters and regions are not easily visible without performing multiple aggregations and filters.

• Business decisions related to inventory planning, retailer partnerships, and pricing strategies are often made without clear data driven insights.

II. Solution

• Clean and validate the dataset by correcting data types, removing inconsistencies, and converting percentage values into numeric format.

• Normalize the dataset into well-structured relational tables including retailers, products, locations, and sales transactions.

• Establish foreign key relationships between tables to ensure consistent and reliable data storage.

• Store transactional information in a central sales table while separating descriptive data into dimension tables.

• Apply advanced SQL techniques such as joins, aggregations, window functions, and CTEs to analyse business performance.

- Address key business questions such as
  - Which retailers generate the highest total sales revenue?
  - Which states and cities generate the highest revenue?
  - Which sales methods contribute the most revenue?
  - Which products generate the highest operating profit?
  - Are sales increasing or decreasing across quarters?
  - Which retailers perform best within each state?

Dataset Description

Dataset Name: Nike Sales Data

The dataset contains transactional retail sales information including product sales, retailer information, location data, and financial metrics.

Approximate dataset size: 10000 plus records

Key columns include

- Retailer
- Retailer_ID
- Invoice_Date
- Sales_Quarter
- State
- City
- Product
- Price_per_Unit
- Units_Sold
- Sales_Method
- Total_Sales
- Operating_Cost
- Operating_Profit
- Operating_Margin

These attributes allow analysis of sales revenue, profitability, product demand, and geographic performance.

Data Cleaning and Preparation

Several preprocessing steps were performed before analysis.

Removed duplicate records.
Standardized data types such as dates and numeric values.
Converted percentage operating margin values into numeric format.
Trimmed inconsistent text values.
Ensured all foreign key relationships matched across tables.

Operating margin values originally stored as text percentages were converted using SQL transformation.

Example: - REPLACE (Operating_Margin,'%',''):::DECIMAL

This conversion enables accurate financial calculations.

Database Normalization

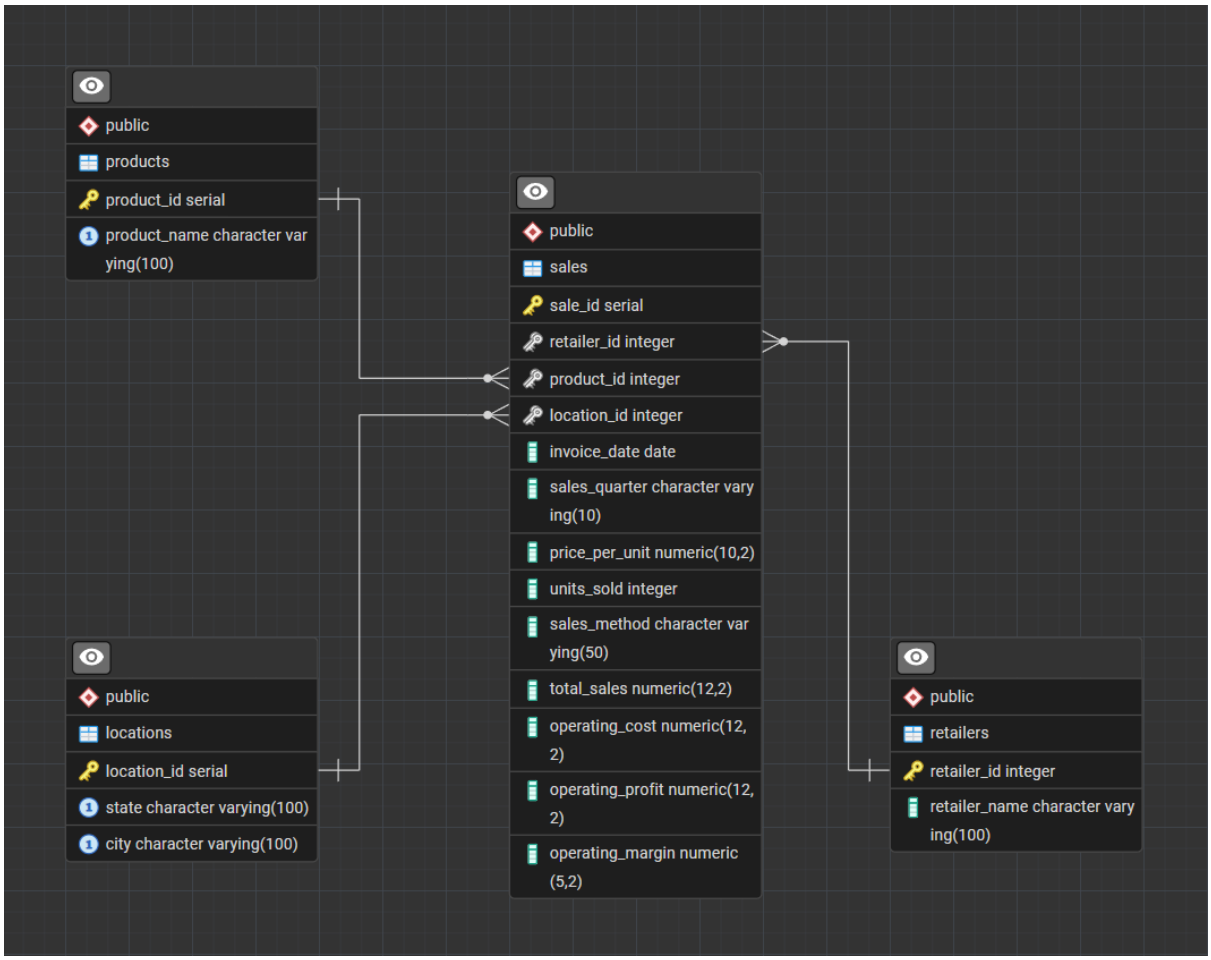
The original dataset contained repeated information across rows. This leads to redundancy and inefficient storage.

The dataset was normalized into four relational tables.

- Retailers Table - Stores unique retailer information.
- Locations Table - Stores state and city combinations.
- Products Table - Stores product names.
- Sales Table - Stores transactional sales data and references other tables through foreign keys.

This structure improves query performance and ensures data consistency.

Database Schema and Relationships



The normalized database follows a relational model where the sales table acts as the central fact table.

Relationships include

Retailers to Sales
One retailer can have multiple sales transactions.

Products to Sales
Each product can appear in many sales records.

Locations to Sales
Each location may contain multiple transactions.

This structure enables efficient analysis through SQL joins.

Business Analysis Using SQL

The analysis of the Nike retail sales dataset is divided into three major categories to evaluate retailer performance, product demand, and overall sales trends. SQL queries were used to extract insights from the normalized relational database.

Category 1: Retailer Performance Analysis

1. Identify the retailers generating the highest total sales revenue.

```sql
SELECT r.retailer_name,
SUM(s.total_sales) AS total_revenue
FROM sales s
JOIN retailers r ON s.retailer_id = r.retailer_id
GROUP BY r.retailer_name
ORDER BY total_revenue DESC
LIMIT 5;
```

	retailer_name character varying (100)	total_revenue numeric
1	Amazon	110252007.38
2	Ajio	98884091.88
3	Nike Store	75884096.63
4	Myntra	60222408.69
5	Flipkart	37855216.81

Insight
A small number of retailers contribute a large portion of the overall revenue. These retailers act as key distribution partners for Nike products. Strengthening relationships with these retailers may increase overall sales performance.

2. Identify retailers with the highest operating margin.

```sql
SELECT r.retailer_name,
AVG(s.operating_margin) AS avg_margin
FROM sales s
JOIN retailers r ON s.retailer_id = r.retailer_id
GROUP BY r.retailer_name
ORDER BY avg_margin DESC;
```

	retailer_name character varying (100)	avg_margin numeric
1	Nike Store	17.6028235294117647
2	Ajio	17.5188612099644128
3	Myntra	17.4439953810623557
4	Flipkart	17.4044526901669759
5	Amazon	17.3889980353634578

Insight

Some retailers generate higher profit margins compared to others. These retailers demonstrate more efficient pricing strategies and operational performance.

3. Find the top performing retailer in each state.

```sql
WITH retailer_sales AS (
SELECT l.state,
r.retailer_name,
SUM(s.total_sales) AS total_revenue
FROM sales s
JOIN retailers r ON s.retailer_id = r.retailer_id
JOIN locations l ON s.location_id = l.location_id
GROUP BY l.state, r.retailer_name
),
ranked_sales AS (
SELECT state,
retailer_name,
total_revenue,
RANK() OVER (PARTITION BY state ORDER BY total_revenue DESC) AS rank_position
FROM retailer_sales
)
SELECT state,
retailer_name,
total_revenue
FROM ranked_sales
WHERE rank_position = 1;
```

	state character varying (100)	retailer_name character varying (100)	total_revenue numeric
1	Delhi	Amazon	22788731.93
2	Karnataka	Amazon	22093204.43
3	Kerala	Amazon	22316638.72
4	Maharashtra	Amazon	22877703.34
5	Tamil Nadu	Amazon	20175728.96

Insight

Retailer performance varies across states. Certain retailers dominate specific regional markets. Understanding regional leaders helps improve distribution strategies and partnership planning.

Category 2: Product Performance Analysis

4. Determine the best-selling products by units sold.

```sql
SELECT p.product_name,
SUM(s.units_sold) AS total_units
FROM sales s
JOIN products p ON s.product_id = p.product_id
GROUP BY p.product_name
ORDER BY total_units DESC;
```

	product_name character varying (100)	total_units bigint
1	Jacket	11403
2	T Shirt	11029
3	Running Shoes	10793
4	Sneakers	10698
5	Shorts	10695

Insight

Products with the highest number of units sold indicate strong consumer demand. These products should receive priority in inventory planning and marketing campaigns.

5. Find the most profitable products based on operating profit.

```sql
SELECT p.product_name,
SUM(s.operating_profit) AS total_profit
FROM sales s
JOIN products p ON s.product_id = p.product_id
GROUP BY p.product_name
ORDER BY total_profit DESC;
```

	product_name character varying (100)	total_profit numeric
1	Jacket	13701668.55
2	T Shirt	13573469.47
3	Running Shoes	13183367.91
4	Shorts	13159898.64
5	Sneakers	13008910.26

Insight

Some products contribute significantly to overall profitability. These products play an important role in revenue growth and financial performance.

6. Identify products with high sales but low profit margins.

```sql
SELECT p.product_name,
SUM(s.total_sales) AS total_revenue,
AVG(s.operating_margin) AS avg_margin
FROM sales s
JOIN products p ON s.product_id = p.product_id
GROUP BY p.product_name
HAVING AVG(s.operating_margin) < (
SELECT AVG(operating_margin) FROM sales
)
ORDER BY total_revenue DESC;
```

	product_name character varying (100)	total_revenue numeric	avg_margin numeric
1	Jacket	79774089.87	17.2966887417218543
2	Sneakers	75377900.36	17.3018060836501901

Insight

Certain products achieve strong sales volume but deliver weaker margins. These products may require pricing adjustments or cost optimization to improve profitability.

Category 3: Sales and Market Trend Analysis

7. Analyse sales performance by state and city.

```
SELECT l.state,
l.city,
SUM(s.total_sales) AS total_revenue
FROM sales s
JOIN locations l ON s.location_id = l.location_id
GROUP BY l.state, l.city
ORDER BY total_revenue DESC;
```

	state character varying (100)	city character varying (100)	total_revenue numeric
1	Maharashtra	Mumbai	80101864.05
2	Delhi	Delhi	78559709.47
3	Karnataka	Bangalore	75335026.54
4	Kerala	Kochi	74995836.27
5	Tamil Nadu	Chennai	74105385.06

Insight

Sales performance varies significantly between states and cities. High revenue regions indicate strong customer demand and effective retail presence.

8. Compare sales performance across different sales methods.

```
SELECT s.sales_method,
SUM(s.total_sales) AS total_revenue,
SUM(s.units_sold) AS total_units
FROM sales s
GROUP BY s.sales_method
ORDER BY total_revenue DESC;
```

	sales_method character varying (50)	total_revenue numeric	total_units bigint
1	Online	193522032.63	27494
2	Store	189575788.76	27124

Insight

Different sales methods contribute varying levels of revenue. Understanding which method performs best helps optimize sales strategy.

9. Determine quarterly sales trends.

```
SELECT sales_quarter,
SUM(total_sales) AS total_revenue
FROM sales
GROUP BY sales_quarter
ORDER BY sales_quarter;
```

	sales_quarter character varying (10)	total_revenue numeric
1	Q1	62097701.29
2	Q2	133772007.60
3	Q3	60547564.04
4	Q4	126680548.46

Insight

Quarterly analysis reveals seasonal variations in sales performance. Certain quarters show higher demand, which helps guide inventory planning and promotional strategies.

10. Find cities with declining sales compared to the previous quarter.

```
WITH quarterly_sales AS (
SELECT l.city,
s.sales_quarter,
SUM(s.total_sales) AS revenue
FROM sales s
JOIN locations l ON s.location_id = l.location_id
GROUP BY l.city, s.sales_quarter
),
sales_comparison AS (
SELECT city,
sales_quarter,
revenue,
LAG(revenue) OVER (PARTITION BY city ORDER BY sales_quarter) AS previous_revenue
FROM quarterly_sales
)
SELECT city,
sales_quarter,
revenue,
previous_revenue
FROM sales_comparison
WHERE revenue < previous_revenue;
```

	city character varying (100)	sales_quarter character varying (10)	revenue numeric	previous_revenue numeric
1	Bangalore	Q3	10517620.72	24910867.47
2	Chennai	Q3	11910789.62	25797317.02
3	Delhi	Q3	12568046.38	27425515.66
4	Kochi	Q3	11984412.59	26971452.84
5	Mumbai	Q3	13566694.73	28666854.61

Insight

Declining sales in specific cities indicate potential market challenges. These regions may require targeted marketing campaigns or promotional activities to improve performance.

This category-based analysis helps the business understand retailer contribution, product demand, and regional sales patterns. The insights support better decision making in areas such as inventory management, pricing strategy, retailer partnerships, and regional marketing initiatives.

Key Insights

A small number of retailers generate most of the sales revenue.

Certain products dominate total unit sales, indicating strong consumer demand.

Sales performance varies significantly across states and cities.

Some products generate high revenue but lower profit margins.

Seasonal sales patterns exist across quarters.

Retailer profitability varies depending on pricing and operational efficiency.

Business Recommendations

Prioritize partnerships with high performing retailers.

Increase inventory levels for high demand products.

Investigate cost structure of low margin products.

Develop region specific marketing strategies.

Monitor declining sales regions and implement corrective strategies.

Use quarterly trends to optimize promotional campaigns.

Conclusion

This project demonstrates how SQL can transform raw transactional retail data into meaningful business insights. By cleaning and normalizing the dataset into relational tables, the project created a structured database that supports efficient analysis.

Using SQL techniques such as joins, aggregations, common table expressions, and window functions, the analysis evaluated retailer performance, product demand, regional sales trends, and overall profitability. These insights help businesses make better decisions related to inventory management, pricing strategies, sales planning, and retailer partnerships.

The project highlights the importance of proper database design and SQL based analytics in converting raw retail data into actionable business intelligence that supports strategic decision making.